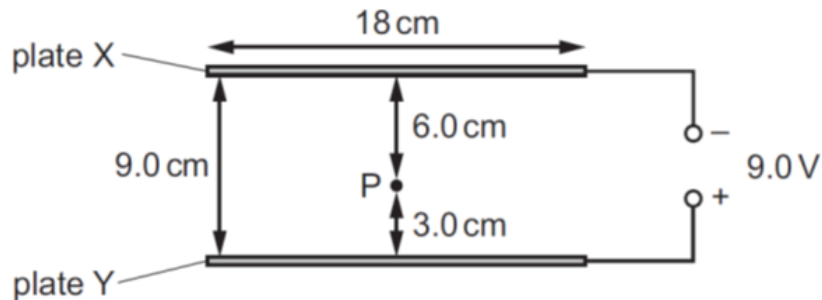


20

Two parallel circular metal plates X and Y, each of diameter 18 cm, have a separation of 9.0 cm. A potential difference of 9.0 V is applied between them.



Point P is 6.0 cm from the surface of plate X and 3.0 cm from the surface of plate Y.

What is the electric field strength at P?

A50 N C⁻¹**B**100 N C⁻¹**C**150 N C⁻¹**D**300 N C⁻¹**[Turn over]**

Answer: B

$$E = V/d = 9.0 / 0.09 = 100 \text{ N C}^{-1}$$

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The electric field strength due to a point charge at a distance r from the charge is E